TCFMAD: Tri-Consistency Foundation Model-Based Few-Shot Manufacturing Anomaly Detection

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***Abstract*—Anomaly detection (AD) aims to distinguish anomalous samples from normal ones and localize the corresponding anomalous regions in real-world manufacturing environments. Benefiting from the strong visual priors of foundation models (FMs), few-shot FM-based methods can rapidly adapt to new manufacturing categories using only a small support set. However, existing methods still suffer from triple inconsistencies: inconsistency between synthetic and real anomalies before training, inconsistency in anomaly evidence across patch tokens from different FM layers during training, and inconsistency between image-level and pixel-level assessments during evaluation. Consequently, AD models may correctly distinguish anomalous samples at the image level while still mislocalizing the anomalous regions at the pixel level. Detection performance further degrades markedly when adapting to other manufacturing datasets under the sparse few-shot setting. To address these inconsistencies, we propose TCFMAD, a tri-consistency FM-based few-shot AD pipeline that enforces consistency in anomaly synthesis, cross-layer anomaly evidence, and fine-grained label construction throughout training and evaluation. TCFMAD adapts the anomaly detector using a few normal support images together with an equally small class-specific anomalous-patch bank. Specifically, realistic anomaly synthesis (RAS) uses this anomalous-patch bank, matched to the 1/2/4-shot normal-support budget, to generate realistic synthetic anomalous samples and alleviate the inconsistency between synthetic and real anomalies. During training, cross-layer evidence agreement (CLEA) optimizes the anomaly predictor with a consistency regularization loss to learn stable anomaly evidence from multi-layer patch tokens of DINOv3. CLEA strengthens the consistency of anomaly evidence extraction. For PKU-Market-PCB, weakly-constrained bootstrapping (WCB) operates offline to refine weak bounding boxes into masks for anomalous-patch extraction and fixed fine-grained localization assessment; it is not part of the anomaly detector or its online inference path. TCFMAD achieves state-of-the-art performance on the human-mask-based public benchmarks and strong relative performance on PKU-Market-PCB under a shared WCB-refined pseudo-mask protocol. Code is available at: https://github.com/GingerCohle/TCFMAD**

***Note to Practitioners*—In real-world manufacturing inspection, anomaly detection models often need to adapt to a new product category using highly limited support data while still providing reliable defect localization for quality control. In practice, existing few-shot anomaly detection methods may report strong image-level performance, yet still produce unstable or misleading localization results because the synthesized anomalies used for training are unrealistic, anomaly evidence varies across different foundation-model layers, and dense pixel-level annotations are often unavailable for evaluation. This article presents a tri-consistency framework to address these practical issues at three stages of deployment. Specifically, the proposed method improves the realism of synthetic anomalies before training, stabilizes anomaly evidence across multiple DINOv3 layers during training, and refines fine-grained masks from weak annotations during evaluation. These properties are especially relevant to industrial scenarios such as PCB inspection and other manufacturing tasks where annotation cost is high and rapid adaptation is required. Practitioners should note that TCFMAD is not a strictly normal-only adaptation method. In addition to a few normal reference images, RAS requires an equally small set of class-specific anomalous patches. WCB further uses limited dense annotations and weak bounding boxes offline to extract the PKU-Market-PCB patch bank and construct fixed evaluation masks; it does not increase online detector inference cost. Nevertheless, it provides a practical way to improve both anomaly discrimination and localization reliability in sparse-data manufacturing scenarios.**

***Index Terms*—Few-shot anomaly detection, foundation models, cross-layer consistency, anomaly-evidence predictor.**

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I. Introduction

ANOMALY detection (AD) [1] aims to distinguish anomalous samples from normal ones and localize the corresponding anomalous regions in real-world manufacturing environments. It is a fundamental task in computer vision

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and plays a vital role across diverse manufacturing scenarios, including microelectronics [2], [3], metal processing [4], [5], food manufacturing [6], and pharmaceuticals [7], [8], [9]. In practice, the target product is usually fixed while the associated anomaly categories are diverse and often difficult to enumerate in advance. In manufacturing scenarios, AD models are often expected to adapt rapidly [10] while maintaining high precision and recall for reliable anomaly detection and localization. Moreover, fine-grained annotations for all possible anomaly categories are essential but costly to obtain. To meet these requirements, few-shot AD [11], [12], [13], [14], [15] has emerged as a promising paradigm for rapidly adapting anomaly detection models to new manufacturing scenarios without requiring extensive fine-grained annotations. Few-shot AD aims to adapt a detector to a new product category using a highly limited support set. In the setting considered in this work, each product category provides $I \in \{1, 2, 4\}$ normal support images and an equally small anomalous-patch bank used exclusively by RAS. A well-trained AD model is then commonly evaluated with both image-level and pixel-level metrics, which together provide a more comprehensive assessment of anomaly discrimination and localization performance. Current few-shot AD methods often report strong performance on public multi-class benchmarks [1], [16]. However, when these methods are transferred to proprietary manufacturing datasets, the high AUROC and AUPR values reported on public benchmarks often decrease markedly. This indicates the limited robustness of current few-shot AD methods in real-world scenarios. To better understand this performance drop, we investigate the inconsistencies observed in the AD model before training, during training, and during evaluation. We summarize them as the following triple inconsistencies.

A. Synthetic-to-Real Anomaly Inconsistency

Few-shot AD typically relies on contrastive learning between normal samples and synthetically generated anomalous samples to train the anomaly detector. However, existing anomaly synthesis methods often follow traditional strategies [17] that crop or perturb regions from the same normal image and paste them back. Such synthetic anomalies usually resemble local texture corruption or random pasted patches rather than realistic defects with structural priors and category consistency. In addition, unconstrained pasting may place synthetic anomalies on background or irrelevant regions. As a result, the model tends to learn synthetic noise patterns rather than real anomalous patterns, which leads to unstable generalization in few-shot settings. Therefore, a more realistic anomaly synthesis strategy is needed.

B. Cross-Layer Anomaly Evidence Inconsistency

Recent few-shot AD methods [18], [19] often rely on patch tokens from ViT-based foundation-model encoders such as DINO [20], [21], [22] and CLIP [23] to extract anomaly evidence from a single layer. However, as shown in Fig. 1, the regions highlighted as anomaly evidence by different layers vary and may contain noise. This instability arises because patch-token responses can vary substantially across layers and spatial positions, so a strong hotspot in one layer is often not supported by others. A key reason is that different Transformer layers capture texture and structural cues with different sensitivities [24]. Sparse and noisy supervision in few-shot settings [25] makes anomaly evidence more susceptible to layer-specific accidental activation. Consequently, single-layer evidence is often insufficient for reliable anomaly localization, especially for small anomalies in printed circuit board (PCB) inspection.

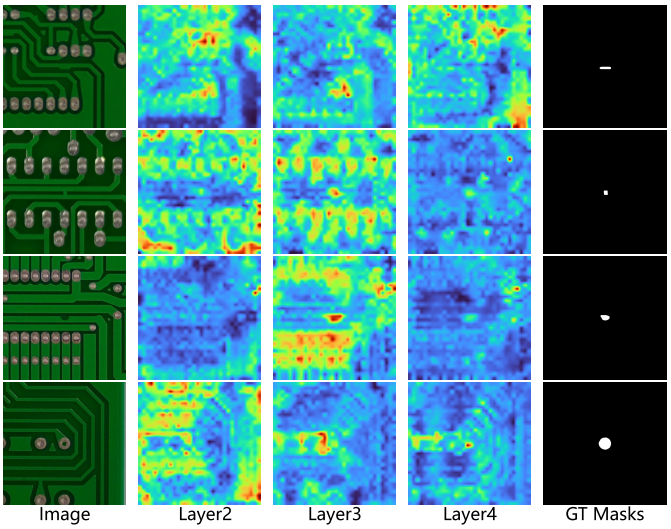


Fig. 1. Examples of heatmaps from different layers. The highlighted potential anomalous regions vary across layers.

C. Image-to-Pixel Assessment Inconsistency

In most manufacturing scenarios, finely annotated datasets are unavailable. Many manufacturing datasets provide only bounding-box, image-level, or other coarse-grained annotations rather than pixel-level anomaly segmentation masks. When only bounding-box annotations are available, a common but problematic practice is to regard every pixel inside a box as anomalous. However, manufacturing defects often occupy only a small portion of the annotated box, while the remaining area contains normal material, background, or regular product structures. Consequently, pixel- and region-level metrics may be affected by box extent and annotation tightness rather than solely reflecting the spatial alignment between predicted anomaly evidence and the actual defect. Directly using coarse boxes as pixel-level References can therefore introduce box-induced localization-assessment bias. Therefore, a fast weakly supervised annotation strategy is needed to refine coarse-grained datasets for evaluation.

Due to the above triple inconsistencies, image-level performance may remain strong even when pixel-level AUROC and PRO degrade, making localization performance difficult to assess reliably. The main challenges are as follows: 1) How can few-shot AD synthesize more realistic anomalous samples for contrastive training? 2) How can cross-layer consistency be enforced when extracting anomaly evidence from multi-layer DINOv3 patch tokens? 3) When fine-grained annotations are scarce, how can foundation models be used to segment anomalous regions and support reliable pixel-level evaluation?

In this paper, we address these inconsistencies by proposing a foundation-model-based few-shot AD method, termed

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TCFMAD, that enforces triple consistency across three stages: before training, during training, and during evaluation. Specifically, realistic anomaly synthesis (RAS) is introduced to generate realistic synthetic anomalous samples from normal samples before training. It constructs reusable few-shot anomalous patches and blends them into normal samples in a mask-aware manner. As a result, scarce anomaly information is transformed into high-quality training samples for contrastive learning, which improves synthetic-to-real anomaly consistency. Cross-layer evidence agreement (CLEA) extracts patch tokens from multiple layers of DINOv3, aggregates them in a weighted manner, and imposes a consistency loss to enforce consistent anomaly-evidence predictions. Hence, anomaly evidence is learned more consistently across layers, which suppresses spurious hotspots and improves the robustness of anomaly detection and localization. To reduce evaluation-label inconsistency, WCB operates offline by lightly fine-tuning the SAM mask decoder on a small mask-annotated anomaly set and then converting weak bounding boxes into a fixed set of fine-grained reference masks. These masks support detector-independent localization assessment without changing detector predictions.

We demonstrate the effectiveness of the proposed method under few-shot settings on the public benchmarks MVTec-AD [1] and VisA [16]. We further adapt TCFMAD to the real-world manufacturing dataset PKU-Market-PCB [26] to verify its practicality in industrial scenarios. State-of-the-art image- and pixel-level results on MVTec-AD and VisA are evaluated using their provided human annotations, whereas the PKU-Market-PCB localization results are reported separately under the shared WCB-refined pseudo-mask protocol.

In summary, the key contributions of this work are as follows:

1) We identify and address a previously overlooked triple inconsistency problem in foundation-model-based few-shot anomaly detection for manufacturing. Specifically, we show that existing methods suffer from synthetic-to-real anomaly inconsistency before training, cross-layer anomaly evidence inconsistency during training, and inconsistency between image-level and pixel-level evaluation. These inconsistencies jointly undermine localization reliability and robustness under sparse supervision in real manufacturing scenarios.
2) Building on the controlled anomaly-reconstruction backbone of FOUNDAD, we propose TCFMAD to mitigate the above inconsistencies across three stages without redesigning the underlying predictor. Before training, RAS constructs realistic synthetic anomalous samples from a class-specific anomalous-patch bank whose size matches the 1/2/4-shot normal-support budget. During training, CLEA extends single-layer reconstruction to learnable multi-layer token aggregation and introduces a cross-layer consistency regularization objective. During evaluation, WCB addresses coarse-annotation-induced assessment bias by refining weak bounding boxes into a fixed set of fine-grained pseudo masks shared by all compared detectors, thereby reducing normal-pixel contamination and enabling more spatially precise localization assessment.
3) Extensive experiments demonstrate the effectiveness and practical value of TCFMAD. On the public few-shot benchmarks MVTec-AD and VisA, TCFMAD achieves state-of-the-art performance in both image-level and pixel-level evaluations under both multi-class and one-class settings. On PKU-Market-PCB, TCFMAD demonstrates strong relative performance under a common WCB-refined pseudo-mask assessment protocol; these localization values are reported separately from the human-ground-truth results on the public benchmarks.

The remainder of this paper is organized as follows. Section II reviews related work and provides the theoretical foundations for this study. Section III details TCFMAD, including realistic anomaly synthesis, cross-layer evidence agreement, and weakly-constrained bootstrapping. The implementation details and experimental results are presented in Section IV. Finally, Section V concludes our work and outlines potential future research directions.

II. Related Work

A. Foundation-Model-Based Anomaly Detection

Recently, vision and language foundation models (FMs) [20], [21], [22], [23], [27] have shown strong performance across a wide range of downstream tasks by leveraging powerful visual and linguistic priors. The DINO series [20], [21], [22] learns highly transferable visual representations from large-scale data, providing a strong feature backbone for few-shot anomaly perception. SAM [27] supports promptable segmentation with remarkable generalization, making it suitable for pseudo-mask generation and refinement under weak supervision. CLIP [23] provides semantic priors through vision-language alignment, which can complement visual representations in anomaly-sensitive understanding under limited data. LLaVA [28] enables multimodal instruction following by connecting a vision encoder with a large language model through visual instruction tuning. GPT-4V extends GPT-4 Vision [29] to process both image and text inputs for general-purpose multimodal understanding and reasoning. Inspired by the above FMs, LogiCode [30] introduces an interpretable framework for logical anomaly detection by using large language models to reason about industrial scenes and automatically generate Python code for anomaly identification. AnomalyGPT [31] addresses threshold-free industrial anomaly detection by adapting LLaVA [28] with simulated anomaly-text data, fine-grained semantic decoding, and prompt learning. PromptAD [11] formulates few-shot anomaly detection as a one-class prompt learning problem based on CLIP [23], where semantic concatenation and an explicit anomaly margin are used to avoid the need for real anomaly images during training. VGFR [12] performs multimodal industrial anomaly detection by reconstructing 3D geometric features from DINO-based 2D visual features, thereby exploiting appearance-geometry consistency for anomaly localization without directly fusing multimodal features. In practice, AD models need to adapt rapidly to a given manufacturing scenario. Consequently, few-shot learning has emerged as a prevalent training paradigm.

B. Few-Shot Anomaly Detection

SPADE [32] introduces anomaly segmentation into nearest-neighbor-based detection by establishing multi-resolution

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correspondences between a test image and similar normal images, achieving effective localization with virtually no training overhead. PatchCore [33] builds a representative memory bank of normal patch features for nearest-neighbor matching, providing strong industrial anomaly detection and localization performance in cold-start settings. AnomalySD [13] adapts stable diffusion [34] to few-shot multi-class industrial anomaly detection through anomaly-region inpainting guided by hierarchical text prompts and mask-based strategies, where reconstruction discrepancies are used for anomaly identification. WinCLIP [18] addresses zero-shot and few-normal-shot anomaly classification and segmentation by aligning multi-scale CLIP features with compositional text prompts, while WinCLIP+ further improves performance by incorporating normal-image references. FastRecon [14] casts few-shot anomaly detection as a support-query alignment problem and introduces a distribution-regularized regression to reconstruct normal query features from a small set of normal references. IIPAD [15] tackles one-for-all few-shot anomaly detection with a vision-language model equipped with a class-shared prompt generator for instance-adaptive prompt learning, together with an improved multi-class memory bank design. FOUNDAD [19] formulates few-shot anomaly detection as a manifold-projection problem in the embedding space of a frozen foundation visual encoder. A lightweight encoder–decoder predictor is trained to reconstruct input patch tokens toward their normal anchors, and the input–reconstruction discrepancy is subsequently used as anomaly evidence. TCFMAD retains this reconstruction backbone to enable a controlled investigation, but addresses three limitations that are not considered in FOUNDAD. Specifically, TCFMAD improves the realism of the synthetic training supervision through RAS, extends single-layer reconstruction to cross-layer representation and consistency optimization through CLEA, and introduces WCB as an independent weak-annotation refinement procedure for reliable localization assessment.

III. Methodology

We consider a sparsely supervised few-shot manufacturing anomaly detection setting. For each product category $c \in C$, the detector-side support data comprise $I \in \{1, 2, 4\}$ normal images, $\mathcal{N}_c = \{X_i^c\}_{i=1}^I$, and an equally small class-specific anomalous-patch bank, $\mathcal{P}_c = \{P_i^c\}_{i=1}^I$. The normal images provide clean anchor features, whereas the anomalous patches are used exclusively by RAS to synthesize anomalous training views. No large-scale anomalous training set is used. During evaluation, normal and anomalous samples are mixed, and the goal is to identify anomalous images and localize their anomalous regions using anomaly evidence.

In addition to the detector-side support data, WCB uses a small auxiliary set with anomaly segmentation masks to adapt the SAM mask decoder. On PKU-Market-PCB, the adapted SAM is applied offline to disjoint weakly box-annotated patch-support and evaluation subsets. Masks from the patch-support subset are used to extract the anomalous patches in $\mathcal{P}_c$, whereas masks from the evaluation subset are fixed and shared by all detectors for localization assessment. WCB is not part of the anomaly-evidence predictor or its online inference path.

Relationship to FOUNDAD. TCFMAD builds on the anomaly-reconstruction backbone used in FOUNDAD [19]. Specifically, we inherit its frozen DINOv3 visual encoder, ViT-based encoder–decoder predictor, normal-anchor reconstruction paradigm, and distance-based anomaly scoring mechanism. These inherited components define the controlled baseline and are not claimed as contributions of this work. TCFMAD extends this backbone at three different stages. Before training, RAS replaces conventional normal-image self-pasting with class-consistent anomalous patches and mask-aware blending. During training, CLEA replaces single-layer reconstruction with learnable multi-layer token aggregation and introduces a cross-layer consistency objective. Offline, WCB refines weak annotations for support-patch extraction and localization assessment while remaining independent of the anomaly-evidence predictor and its online inference path. Therefore, the contribution of TCFMAD lies not in redesigning the reconstruction backbone, but in establishing a tri-consistency pipeline around and within this backbone. Existing few-shot FM-based AD methods often suffer from: (i) synthetic-to-real anomaly inconsistency, (ii) cross-layer anomaly evidence inconsistency, and (iii) image-to-pixel assessment inconsistency. The above inconsistencies occur across three stages: before training, during training, and during evaluation. To address these issues, we propose TCFMAD, a tri-consistency FM-based few-shot AD pipeline that enforces triple consistency as illustrated in Fig. 2 and Fig. 3. Specifically, TCFMAD consists of three components:

- Realistic anomaly synthesis (RAS) generates realistic synthetic anomalies by pasting realistic anomalous patches onto normal training samples X_i^c with mask-aware blending and geometric constraints. Consequently, the gap between synthetic and real anomalies is reduced.
- Cross-layer evidence agreement (CLEA) stabilizes anomaly evidence extraction by aggregating patch tokens from multiple layers of the FM DINOv3 [22]. A consistency regularization loss is further introduced to enforce cross-layer consistency of anomaly evidence.
- Weakly-constrained bootstrapping (WCB) leverages a small set of anomaly segmentation masks to fine-tune the promptable segmentation FM SAM [27] and then bootstraps reliable segmentation masks under weak box supervision.

The details of TCFMAD are presented in the following subsections.

A. Realistic Anomaly Synthesis (RAS)

Although normal-only self-pasting avoids anomalous supervision, the resulting synthetic patterns often resemble arbitrary texture corruption rather than realistic manufacturing defects. TCFMAD therefore introduces a highly limited anomalous-patch bank to provide category-consistent defect morphology and texture priors. Importantly, this anomalous supervision remains few-shot: under the I-shot setting, only I anomalous patches are used per category. Traditional anomaly synthesis methods such as CutPaste [17] generate anomalous samples by cutting a patch from a normal image and pasting it back onto the same image at another location, thereby creating an artificial local irregularity. Although self-pasting is effective,

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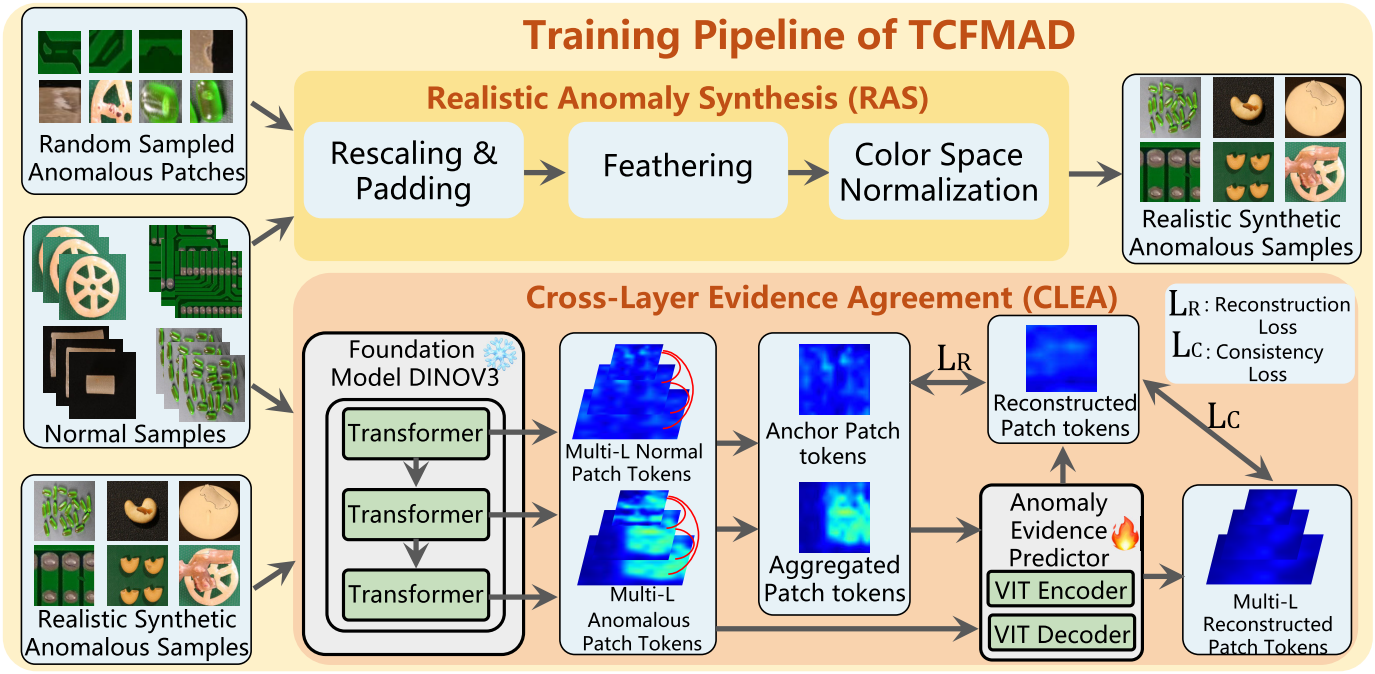


Fig. 2. Overall training pipeline of TCFMAD. The frozen DINOv3 encoder, ViT-based anomaly-evidence predictor, and reconstruction-based anomaly scoring follow the backbone used in FOUNDAD [19]. TCFMAD introduces RAS to construct realistic synthetic anomalous samples before training and CLEA to aggregate multi-layer patch tokens and optimize the predictor with reconstruction and cross-layer consistency losses during training. The WCB evaluation branch is separately illustrated in Fig. 3.

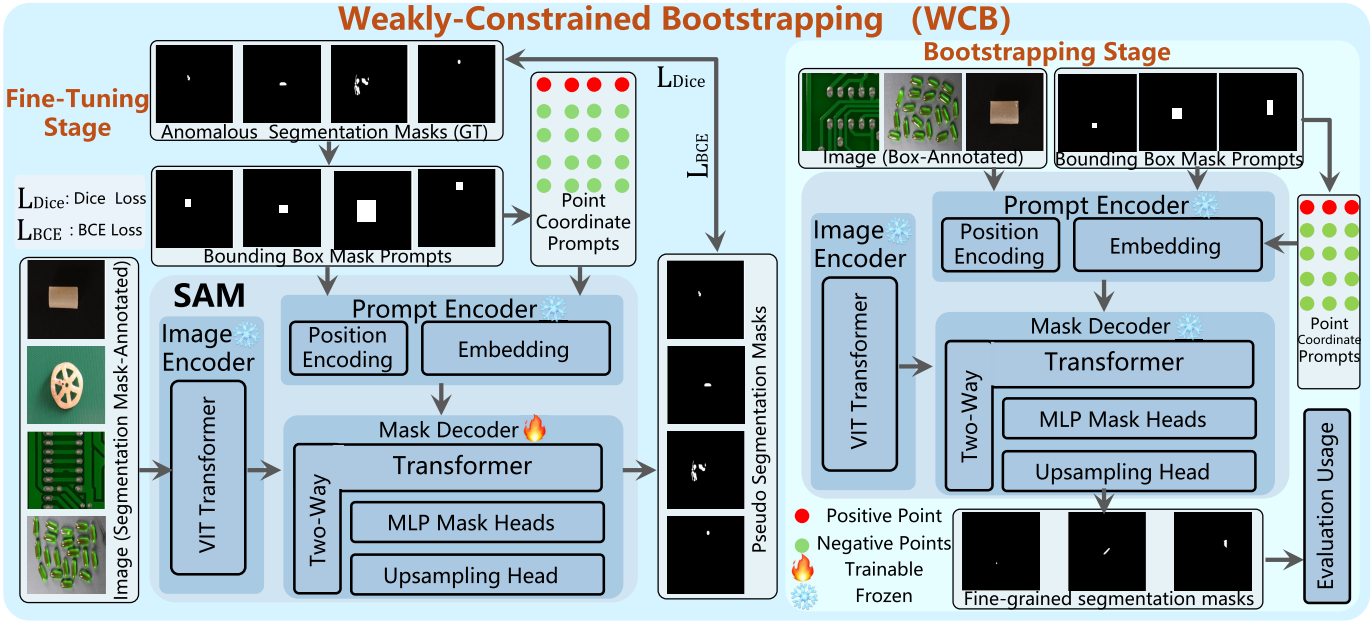


Fig. 3. Overall fine-tuning and bootstrapping pipeline of WCB (Sec. III-C). During fine-tuning, a small set of anomalous images and their corresponding segmentation masks is fed into SAM. The image and prompt encoders are frozen, and only the mask decoder is trainable. The BCE and Dice losses are used to optimize the mask decoder. During bootstrapping, all parameters are frozen, and the fine-tuned SAM is applied to disjoint weakly box-annotated subsets to generate masks for PKU-Market-PCB support-patch extraction and fixed localization assessment.

it often produces synthetic artifacts whose appearance, morphology, and texture differ from those of real manufacturing anomalies, resulting in synthetic-to-real inconsistency. To mitigate this issue, we propose realistic anomaly synthesis (RAS), which synthesizes anomalous training views using real anomalous patches.

Specifically, for each category c, we construct the class-specific anomalous-patch bank $\mathcal{P}_c = \{P_i^c\}_{i=1}^{I}$, which contains exactly I patches and matches the normal-support budget. All patches are sampled within the same product category; cross-category patches are not used. For MVTec-AD and VisA, each patch is extracted from an anomalous support image

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using its provided segmentation mask. For PKU-Market-PCB, WCB first refines the bounding box of a disjoint anomalous support image, after which the masked anomalous region is extracted as an RAS patch. For a normal image X_i^c, RAS first samples an area ratio $r \sim (r_{\min}, r_{\max})$ and rescales a sampled anomalous patch by bilinear interpolation while preserving its aspect ratio, yielding a resized anomalous patch $\overline{P}_c$. To make the patch representation compatible with the normal image $X_i^c \in \mathbb{R}^{3\times H\times W}$, each anomalous patch $\overline{P}_c$ is padded to the spatial size of X_i^c. Pixels outside the anomalous region are set to $(0, 0, 0)$ in the RGB channels. This class-specific anomalous patch avoids the overly strong assumption that different target products share identical defect morphology and texture characteristics. A target location is then sampled from all valid positions such that $\overline{P}_c$ lies entirely within the image boundary. To avoid hard boundary artifacts, we apply a feathering operation [35] to $\overline{P}_c$ and obtain a soft blending coefficient α. Before blending, both $\overline{P}_c$ and X_i^c are normalized so that they reside in the same color space. Finally, the anomalous view is synthesized by alpha blending:

$$\overline{X}_i^c = (1-\alpha)\odot X_i^c + \alpha \odot \overline{P}_c, \tag{1}$$

where $\odot$ denotes element-wise multiplication and α is broadcast to three channels. The number of anomalous patches in $\mathcal{P}_c$ is exactly I, the same as the number of normal support images in $\mathcal{N}_c$.

Overall, RAS narrows the synthetic-to-real gap by generating class-consistent and realistic anomalies from the limited normal support images. As a result, it provides stronger and more reliable synthetic anomalous training samples $\overline{X}_i^c$ under extreme few-shot imbalance.

B. Cross-Layer Evidence Agreement (CLEA)

After anomaly synthesis, both the synthetic anomalous samples $\overline{X}_i^c$ and the normal samples X_i^c are fed into the foundation model DINOv3 [22] to obtain patch tokens for anomaly evidence extraction. The anomaly evidence is essential for distinguishing anomalous samples and localizing anomalous regions. However, existing AD methods [19] rely on single-layer patch tokens, making the resulting anomaly evidence layer-dependent [36] and unstable. Spurious hotspots may appear only in a single layer and thus produce misleading anomaly evidence, especially under the sparse few-shot setting. To stabilize anomaly evidence, we propose cross-layer evidence agreement (CLEA), which enforces anomaly agreement across multiple layers via weighted fusion and a consistency regularization loss. CLEA trains an anomaly-evidence predictor Ψ with the frozen DINOv3 encoder Φ^* to distinguish anomalous samples from normal ones and localize the corresponding anomalous regions. We extract patch tokens from the second-, third-, and fourth-last Transformer blocks, corresponding to Blocks 9–11 under one-based indexing. For a 512×512 input, each block provides 1,024 patch tokens with a feature dimension of 768. The `[CLS]` and register tokens are excluded, and all three layers are obtained in a single backbone forward pass during both training and inference. We describe below how CLEA facilitates both training and inference.

Inspired by empirical observations, patch tokens from normal samples in DINOv3 tend to be closer to one another in the feature space than to those from anomalous samples [19], which is consistent with its contrastive learning paradigm. Hence, we utilize frozen DINOv3 Φ^* to extract patch tokens from multiple layers for both normalsamples $X_i^c \in \mathbb{R}^{3\times H\times W}$ and synthetic anomalous samples $\overline{X}_i^c \in \mathbb{R}^{3\times H\times W}$ as follows:

$$\begin{aligned} Z_{i,j}^c &= \Phi^*(X_i^c), \\ \overline{Z}_{i,j}^c &= \Phi^*(\overline{X}_i^c), \end{aligned} \tag{2}$$

where $Z_{i,j}^c$ and $\overline{Z}_{i,j}^c$ denote the multi-layer patch tokens extracted from normal and synthetic anomalous samples, respectively. Here, j indexes the layer of DINOv3, and $\overline{Z}_{i,j}^c$ denotes the token at layer j in $\overline{Z}_i^c$. The superscript $*$ indicates that all parameters are kept frozen. The normal patch tokens $Z_{i,j}^c$ are used as anchor tokens, whereas the anomalous patch tokens $\overline{Z}_{i,j}^c$ are contrasted with $Z_{i,j}^c$ to localize anomalous regions through their latent anomaly evidence. However, in real manufacturing scenarios, such anchor patch tokens are not directly available for each test sample at inference time, which makes it difficult to determine whether the sample is anomalous. Therefore, following FOUNDAD [19], we train an anomaly-evidence predictor Ψ to reconstruct the input patch tokens toward the anchor patch tokens. Once training is completed, the feature distance between the input patch tokens and the output patch tokens of the anomaly-evidence predictor Ψ^* is used as anomaly evidence during inference. We refer to the output of Ψ^* as the reconstructed patch tokens. Intuitively, for a normal sample X_i^c, no anomaly evidence is present, so the reconstructed patch tokens $\Psi^*(Z_i^c)$ should remain close to the anchor patch tokens Z_i^c. In contrast, for an anomalous sample $\overline{X}_i^c$, $\Psi^*(\overline{Z}_i^c)$ is expected to suppress anomaly evidence and move toward the anchor patch tokens Z_i^c. As a result, the feature distance between the input anomalous patch tokens $\overline{Z}_i^c$ and the reconstructed patch tokens can be regarded as anomaly evidence. To train a stable anomaly predictor Ψ, we utilize patch tokens from multiple layers rather than a single layer, as follows:

$$\begin{aligned} &\alpha^h = \mathrm{softmax}_j(W_h), \qquad \alpha^z = \mathrm{softmax}_j(W_z), \\ &\overline{P}_{i,\ell}^c = \Psi\left(\sum_j \alpha_j^h \odot \overline{Z}_{i,j}^c\right), \\ &P_{i,\ell}^c = \sum_j \alpha_j^z \odot Z_{i,j}^c, \end{aligned} \tag{3}$$

where $\overline{P}_{i,\ell}^c$ denotes the reconstructed patch tokens obtained from multi-layer anomalous patch tokens containing anomaly evidence, and $P_{i,\ell}^c$ denotes the aggregated anchor patch tokens obtained via weighted summation. $W_h, W_z \in \mathbb{R}^{3\times768}$ are independent learnable fusion matrices for the synthetic context and normal-anchor features, respectively; $\odot$ denotes element-wise multiplication. Both matrices are initialized to zero and normalized by a softmax over the layer dimension j. Thus, the initial fusion is uniform, while each feature dimension can learn a different layer preference. The two matrices are optimized jointly with Ψ and introduce 4,608 trainable parameters. We then optimize Ψ to push the reconstructed patch tokens $\overline{P}_{i,\ell}^c$ toward the anchor patch tokens $P_{i,\ell}^c$. The

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corresponding reconstruction loss is defined using the mean squared error (MSE) loss as follows:

$$\mathcal{L}_r = \sum_c \sum_i \left\| \overline{P}_i^c - P_{i,\ell}^c \right\|_2^2, \tag{4}$$

where the reconstruction loss $\mathcal{L}_r$ encourages the anomaly-evidence predictor Ψ to reconstruct anomalous patch tokens toward their normal anchors. Moreover, to enforce anomaly evidence agreement across layers, we additionally compute per-layer reconstructed patch tokens from $\overline{Z}_i^c$ as follows:

$$\overline{P}_{i,j}^c = \Psi(\overline{Z}_{i,j}^c), \tag{5}$$

where j denotes the layer index. Finally, we impose a consistency loss to align the reconstructed patch tokens from all three layers with the aggregated patch tokens $\overline{P}_{i,\ell}^c$:

$$\mathcal{L}_c = \sum_c \sum_i \sum_j \left\| P_{i,j}^c - \overline{P}_{i,\ell}^c \right\|_2^2. \tag{6}$$

Finally, the overall training objective is defined by the reconstruction loss $\mathcal{L}_r$ and the consistency loss $\mathcal{L}_c$ as follows:

$$\mathcal{L} = \mathcal{L}_r + \lambda \mathcal{L}_c, \tag{7}$$

where $\mathcal{L}_r$ denotes the reconstruction loss and $\mathcal{L}_c$ denotes the consistency loss. The reconstruction loss $\mathcal{L}_r$ encourages the anomaly-evidence predictor to reconstruct anomalous patch tokens toward their normal anchors, while the consistency loss $\mathcal{L}_c$ enforces cross-layer consistency in anomaly evidence.

It is important to note that $\mathcal{L}_c$ does not directly align or homogenize the raw DINOv3 features $\{Z_{i,j}^c\}_j$. The DINOv3 encoder remains frozen, and its layer-specific texture, structural, and semantic cues are preserved. Instead, $\mathcal{L}_c$ regularizes only the reconstructed patch tokens produced by the shared anomaly-evidence predictor. The multi-layer features are first incorporated into the fused representation, after which the consistency objective discourages reconstructed responses that are strongly activated in only one layer but unsupported by the remaining layers. Consequently, CLEA aims to suppress accidental layer-specific evidence rather than remove complementary information from the frozen foundation-model representations.

Anomaly Scoring. At inference, a test image is encoded by the frozen DINOv3 backbone, and the selected multi-layer patch tokens are aggregated to obtain the input representation $\mathbf{Z} \in \mathbb{R}^{N \times D}$, where N is the number of patches and D is the feature dimension. The frozen anomaly-evidence predictor produces reconstructed patch tokens $\widehat{\mathbf{Z}} = \Psi^*(\mathbf{Z})$. The anomaly evidence of patch p is computed as the mean squared feature distance between its input and reconstructed tokens:

$$a_p = \frac{1}{D} \sum_{d=1}^{D} \left(Z_{p,d} - \widehat{Z}_{p,d} \right)^2. \tag{8}$$

A larger a_p indicates a greater deviation from the learned normal representation. The image-level anomaly score is computed by top-K average pooling:

$$s_{\text{img}} = \frac{1}{K} \sum_{p \in \text{TopK}(\mathbf{a}, K)} a_p, \tag{9}$$

where $\mathbf{a} = \{a_p\}_{p=1}^N$ and $\text{TopK}(\mathbf{a}, K)$ returns the indices of the K largest patch-level responses. We set $K = 10$ for MVTec-AD and $K = 6$ for VisA and PKU-Market-PCB. Thus, the image-level score is a top-K average rather than a single maximum response. For pixel-level localization, the $N = 1{,}024$ responses are reshaped into a 32×32 anomaly-evidence map and bilinearly upsampled to 512×512 using `align_corners=false`. No Gaussian smoothing, additional spatial max pooling, or morphological post-processing is applied.

The trained AD model must then be evaluated on test data. However, in many real manufacturing scenarios, well-annotated labels are unavailable, especially anomaly segmentation masks required for localization evaluation. Therefore, in the following subsection, we introduce a foundation-model-based method to generate high-quality anomaly segmentation masks.

C. Weakly-Constrained Bootstrapping (WCB)

Anomaly segmentation masks are often unavailable in manufacturing due to the high cost of annotation. In practice, anomalous samples are usually annotated only with coarse bounding-box supervision, which leads to a mismatch between image-level and pixel-level evaluations. Unlike RAS and CLEA, WCB is not introduced to modify the anomaly detector. Instead, it addresses the evaluation problem that arises when the entire bounding box is used as a pixel-level reference. Such a box labels all enclosed pixels as anomalous even though a substantial portion may correspond to normal background or regular product structures. This annotation mismatch can bias P-AUROC and PRO and make localization assessment dependent on box tightness. To mitigate this issue, we propose weakly-constrained bootstrapping (WCB), which leverages a small set of anomaly segmentation masks to fine-tune the promptable segmentation foundation model SAM [27] and then bootstraps reliable segmentation masks from a weakly box-annotated dataset.

1) Role of WCB: WCB is an offline segmentation and annotation-refinement procedure and is not part of the anomaly-evidence predictor or its online inference path. On PKU-Market-PCB, the fine-tuned SAM mask decoder is applied to two disjoint weakly box-annotated subsets. Masks from the patch-support subset are used to extract the class-specific anomalous patches required by RAS. Masks from the evaluation subset are fixed after bootstrapping and shared by all compared detectors for spatially precise localization assessment. These evaluation masks do not alter detector parameters, predicted anomaly maps, or image-level anomaly scores.

To reduce overfitting with limited data while preserving general segmentation priors, we freeze the image and prompt encoders of SAM and update only its mask decoder during fine-tuning. Let $\tau(\cdot;\theta)$ denote SAM, with parameters $\theta = (\theta_{\text{img}}, \theta_{\text{prm}}, \theta_{\text{dec}})$ corresponding to the image encoder, prompt encoder, and mask decoder, respectively. We keep $(\theta_{\text{img}}, \theta_{\text{prm}})$ frozen and update only θ_{dec}. WCB utilizes two data sources: a small set $\{S_k, M_k\}_{k=1}^K$ with anomaly segmentation masks for fine-tuning, where S_k is an anomalous image and $M_k \in \{0,1\}^{H \times W}$ is its anomaly segmentation mask, and a larger weakly box-annotated set $\{S_t, B_t\}_{t=1}^T$ for bootstrapping, where $B_t \in \{0,1\}^{H \times W}$ is a box mask indicating a coarse anomalous region. The goal is to bootstrap a fine-grained anomaly

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segmentation mask for each box-annotated anomalous image in the weak set such that the resulting mask remains sufficiently compact and noise-free for reliable evaluation.

During fine-tuning, the input prompts for SAM are constructed as follows. A tight box is derived from the anomaly segmentation mask M_k and then dilated using a 3×3 kernel to obtain the box prompt $\overline{B}_k$. The point prompt $\overline{D}_k$ contains one positive point placed near the centroid of the box mask and four negative points selected deterministically from a 12-pixel ring outside the box. The point and box prompts encourage SAM $\tau(\cdot;\theta)$ to focus on the anomalous region while suppressing expansion into the surrounding background. SAM then outputs multiple candidate pseudo-segmentation masks based on the input image and prompts as follows:

$$\overline{M}_k = \text{Binary}(\tau(S_k, \overline{B}_k, \overline{D}_k; \theta)), \tag{10}$$

where SAM first outputs a probability map, which is binarized to obtain the pseudo-segmentation mask $\overline{M}_k \in [0,1]^{H\times W}$. Here, Binary$(\cdot)$ denotes the binary thresholding operation. The mask decoder θ_{dec} is subsequently fine-tuned by minimizing a standard segmentation objective as follows:

$$\text{L}_{ft} = \mathcal{L}_{\text{BCE}}(\overline{M}_k, M_k) + \mathcal{L}_{\text{Dice}}(\overline{M}_k, M_k), \tag{11}$$

where $\mathcal{L}_{\text{BCE}}$ and $\mathcal{L}_{\text{Dice}}$ are the binary cross-entropy and Dice losses, respectively. After fine-tuning, all parameters of SAM $\tau(\cdot;\theta^*)$ are frozen for the subsequent bootstrapping stage.

During bootstrapping, the weak bounding box is likewise dilated with a 3×3 kernel and used as the box prompt. The point prompt D_t contains one positive point near the centroid of the box mask and four deterministic negative points from a 12-pixel ring outside the box. No prediction from any anomaly detector is used to construct these prompts. Consequently, the generated masks are independent of the detector being evaluated. The fine-tuned SAM $\tau(\cdot;\theta^*)$ outputs candidate anomaly segmentation masks as follows:

$$\tilde{M}_t = \text{Binary}(\tau(S_t, B_t, D_t; \theta^*)), \tag{12}$$

where $\tilde{M}_t$ is the fine-grained segmentation mask obtained after bootstrapping. Among the candidate masks, we retain the mask with the highest SAM-predicted score. Its logits are passed through a sigmoid, intersected with the dilated box mask, and binarized using a threshold of 0.5. The implementation does not use an additional area-ratio constraint, iterative prompt tightening, connected-component filtering, or a Top-K fallback.

Overall, WCB provides a practical localization-assessment protocol for manufacturing datasets without human-annotated pixel-level masks. For patch-support images, its masks enable anomalous-patch extraction for RAS; for evaluation images, they serve as fixed fine-grained reference masks shared by all detectors. WCB does not alter detector predictions. Instead, it reduces the normal-pixel contamination introduced by coarse boxes and enables the same anomaly maps to be assessed using more spatially precise reference annotations. The resulting masks remain pseudo annotations and are therefore reported separately from the human-ground-truth masks of MVTec-AD and VisA.

IV. Experiments

A. Experimental Setup

1) Datasets: Following the protocols of IIPAD [15] and FOUNDAD [19], we evaluate TCFMAD on the public few-shot benchmarks MVTec-AD [1] and VisA [16], and further validate its effectiveness on the real-world manufacturing dataset PKU-Market-PCB [26]. MVTec-AD consists of 5,354 images across fifteen categories, including five texture categories (carpet, grid, leather, tile, and wood) and ten object categories (bottle, cable, capsule, hazelnut, metal nut, pill, screw, toothbrush, transistor, and zipper). It contains 3,629 normal training images, 467 normal test images, and 1,258 anomalous test images with corresponding pixel-level masks. VisA consists of 10,821 images spanning twelve categories, which can be broadly divided into five texture-like categories (chewinggum, fryum, two macaroni categories, and pipe fryum) and seven object categories (candle, capsules, cashew, and four PCB categories). It contains 8,659 normal training images, 962 normal test images, and 1,200 anomalous test images with corresponding pixel-level masks. The PKU-Market-PCB dataset was originally designed for anomalous object detection with weak box annotations. We apply the proposed weakly-constrained bootstrapping to generate fine-grained pseudo-segmentation masks for localization assessment. It consists of 3,543 PCB images across six anomaly categories, including 2,363 normal training images, 590 normal test images, and 590 anomalous test images. It covers six real-world PCB defects, namely missing holes, mouse bites, open circuits, shorts, spurs, and spurious copper. Experimental results on both the public few-shot benchmarks and the real-world manufacturing dataset verify the effectiveness of the proposed method.

For MVTec-AD and VisA, P-AUROC and PRO are computed using the provided human-annotated pixel-level masks. PKU-Market-PCB provides only bounding-box annotations. Directly using the corresponding box masks would label substantial normal regions as anomalous and lead to a coarse, box-dependent localization assessment. We therefore use WCB to generate one fixed set of refined pseudo-segmentation masks for all compared detectors. No anomaly prediction from FOUNDAD or TCFMAD is used during mask construction. The resulting PKU-Market-PCB localization metrics are explicitly reported as pseudo-mask-based assessment and are not treated as directly equivalent to the human-ground-truth results on MVTec-AD and VisA.

2) Supervision Protocol: Under the I-shot setting, each product category provides I normal support images and exactly I anomalous patches for detector-side adaptation. The normal images provide anchor features, whereas the anomalous patches are accessed only by RAS for synthesis. On PKU-Market-PCB, WCB additionally uses a small auxiliary segmentation-mask set to fine-tune the SAM mask decoder and disjoint weakly box-annotated subsets for support-patch extraction and evaluation-mask construction. The normal support set, anomalous patch bank, WCB fine-tuning set, and final evaluation set are maintained through independent manifests. The evaluation masks are generated once, shared by all compared detectors, and never used to optimize the anomaly-evidence predictor.

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TABLE I

QUANTITATIVE COMPARISON WITH MULTI-CLASS-ONE-MODEL FEW-SHOT AD METHODS ON VISA AND MVTEC-AD UNDER 1/2/4-SHOT SETTINGS. EACH SHOT SETTING IS REPORTED AS A SEPARATE BLOCK. WE REPORT IMAGE-LEVEL AUROC (I-AUROC, %), IMAGE-LEVEL AUPR (AUPR, %), PIXEL-LEVEL AUROC (P-AUROC, %), AND PRO-AUC (%). THE BEST AND SECOND-BEST RESULTS WITHIN EACH SHOT BLOCK ARE HIGHLIGHTED IN BLUE AND GREEN

Shot	Method	w/o Texts	VisA				MVTec-AD				Reference
			I-AUROC	AUPR	P-AUROC	PRO	I-AUROC	AUPR	P-AUROC	PRO	
1	PatchCore [33]	✓	58.9	62.8	76.7	64.3	63.7	81.2	83.9	72.7	CVPR′2022
	FastRecon [14]	✓	55.0	72.8	70.7	58.2	51.2	72.6	62.1	60.3	ICCV′2023
	WinCLIP [18]	✗	83.1	85.1	94.6	80.9	92.8	96.5	92.4	83.5	CVPR′2023
	PromptAD [11]	✗	80.8	83.2	96.3	82.2	86.3	93.4	91.8	83.6	CVPR′2024
	AnomalySD [13]	✗	86.1	89.1	96.5	93.9	93.6	96.9	94.8	89.2	CVIU′2026
	IIPAD [15]	✗	85.4	87.5	96.9	87.3	94.2	97.2	96.4	89.8	ICLR′2025
	FOUNDAD [19]	✓	92.6	92.0	99.7	98.0	96.1	97.9	96.8	92.8	ICLR′2026
	TCFMAD	✓	93.4	93.3	99.7	98.2	95.8	97.0	97.3	93.6	
2	PatchCore [33]	✓	60.2	64.3	82.4	68.1	72.4	86.2	89.6	74.2	CVPR′2022
	FastRecon [14]	✓	58.2	74.6	79.6	63.5	51.7	74.9	62.4	59.9	ICCV′2023
	WinCLIP [18]	✗	83.7	84.9	95.1	81.8	92.7	96.3	92.4	83.9	CVPR′2023
	PromptAD [11]	✗	84.3	87.8	96.9	84.7	89.2	94.8	92.2	84.3	CVPR′2024
	AnomalySD [13]	✗	87.4	90.1	96.8	94.1	94.8	97.0	95.8	90.4	CVIU′2026
	IIPAD [15]	✗	86.7	88.6	97.2	87.9	95.7	97.9	96.7	90.3	ICLR′2025
	FOUNDAD [19]	✓	93.5	93.0	99.7	98.0	96.8	98.3	97.0	93.3	ICLR′2026
	TCFMAD	✓	94.9	94.5	99.7	98.5	95.7	97.2	97.4	93.6	
4	PatchCore [33]	✓	62.6	69.9	85.4	70.6	74.9	88.8	92.6	80.8	CVPR′2022
	FastRecon [14]	✓	57.6	73.7	78.8	62.9	50.8	73.1	65.0	62.8	ICCV′2023
	WinCLIP [18]	✗	84.1	86.1	95.2	82.1	94.0	96.9	92.9	84.4	CVPR′2023
	PromptAD [11]	✗	85.7	88.8	97.2	84.7	90.6	96.5	92.4	84.6	CVPR′2024
	AnomalySD [13]	✗	88.9	90.9	97.5	94.3	95.6	97.6	96.2	90.8	CVIU′2026
	IIPAD [15]	✗	88.3	89.6	97.4	88.3	96.1	98.1	97.0	91.2	ICLR′2025
	FOUNDAD [19]	✓	94.4	94.0	99.7	98.4	97.1	98.6	97.2	93.5	ICLR′2026
	TCFMAD	✓	95.3	94.7	99.8	98.7	96.1	98.0	96.3	93.0	

3) Evaluation Metrics: To comprehensively evaluate both anomaly detection and localization performance, we adopt three standard metrics: (1) the area under the receiver operating characteristic curve (AUROC), including image-level AUROC (I-AUROC) and pixel-level AUROC (P-AUROC), which measure the separability between normal and anomalous samples at the image and pixel levels, respectively; (2) the area under the precision-recall curve (AUPR), which is particularly informative under class imbalance by emphasizing the trade-off between precision and recall; and (3) the per-region overlap (PRO), which evaluates region-level localization quality by measuring the overlap between predicted anomalous regions and ground-truth regions. Following standard practice in industrial anomaly detection, curve areas involving the false-positive rate are computed only up to an FPR of 0.3 [1]. Together, these metrics provide a comprehensive evaluation of both detection accuracy and localization effectiveness.

4) Comparison Methods: We compare TCFMAD with state-of-the-art anomaly detection methods under both multi-class and one-class settings. For the multi-class-one-model few-shot setting, we compare with PatchCore [33], FastRecon [14], WinCLIP [18], PromptAD [11], AnomalySD [13], IIPAD [15], and FOUNDAD [19], as summarized in Table I. For the one-class-one-model setting, we further compare with classical and recent one-class methods, including PatchCore [33], WinCLIP [18], InCTRL [37], AnomalyCLIP [38], PromptAD [11], and LogSAD [39]. Meanwhile, FOUNDAD [19] and our TCFMAD are evaluated under the more challenging generalized multi-class-one-model few-shot setting, as shown in Table II.

5) Implementation Details: All datasets use 512×512 RGB inputs and ImageNet normalization. FOUNDAD [19] serves as the baseline. For a fair comparison, patch tokens are extracted using the frozen DINOv3 ViT-B/16 model pretrained on LVD-1689M, and a trainable ViT-based encoder–decoder is used as the anomaly-evidence predictor. The predictor, inherited from FOUNDAD, projects 768-dimensional tokens into a 384-dimensional hidden space, applies six Transformer blocks with 12 attention heads and an MLP ratio of 4, and projects the output back to 768 dimensions. It uses neither a class token nor a predictor positional embedding. The predictor and the CLEA fusion matrices are optimized with AdamW using a learning rate of 10^{-3}, betas $(0.9, 0.999)$, an epsilon of 10^{-8}, and a weight decay of 10^{-4}. The learning rate is constant without warm-up. Training is conducted in FP32 without gradient accumulation or gradient clipping. We set $\lambda = 0.1$, and the context branch selects the clean or synthetic

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TABLE II

COMPARISON WITH ONE-CLASS-ONE-MODEL FEW-SHOT AD METHODS ON VISA AND MVTEC-AD. FOUNDAD AND TCFMAD ARE EVALUATED UNDER THE MORE CHALLENGING GENERALIZED MULTI-CLASS-ONE-MODEL FEW-SHOT SETTING. RESULTS ARE REPORTED UNDER THREE SHOT SETTINGS (1-SHOT, 2-SHOT, AND 4-SHOT), WHERE EACH SHOT SETTING IS PRESENTED IN A SEPARATE ROW BLOCK. METRICS INCLUDE IMAGE-LEVEL AUROC (%), AUPR (%), PIXEL-LEVEL AUROC (P-AUROC, %), AND PRO (%). THE BEST AND SECOND-BEST RESULTS WITHIN EACH SHOT SETTING ARE HIGHLIGHTED IN BLUE AND GREEN

Shot	Method	w/o Texts	VisA				MVTec-AD				Reference
			I-AUROC	AUPR	P-AUROC	PRO	I-AUROC	AUPR	P-AUROC	PRO	
1	PatchCore [33]	✓	79.9	82.8	95.4	80.5	83.4	92.2	92.0	79.7	CVPR′2022
	WinCLIP [18]	✗	83.8	85.1	96.4	85.1	93.1	96.5	95.2	87.1	CVPR′2023
	InCTRL [37]	✗	83.2	84.1	89.0	66.7	91.3	95.2	94.6	87.8	CVPR′2024
	AnomalyCLIP [38]	✗	87.7	87.7	83.2	90.1	95.2	97.2	94.6	87.6	ICLR′2024
	PromptAD [11]	✗	86.9	88.4	96.7	85.1	94.6	97.1	95.9	87.9	CVPR′2024
	LogSAD [39]	✗	89.8	90.3	97.5	88.2	95.5	97.3	97.0	92.5	CVPR′2025
	FOUNDAD [19]	✓	92.6	92.0	99.7	98.0	96.1	97.9	96.8	92.8	ICLR′2026
	TCFMAD	✓	93.4	93.3	99.7	98.2	95.8	97.0	97.3	93.6	
2	PatchCore [33]	✓	81.6	84.8	96.1	82.6	86.3	93.8	93.3	82.3	CVPR′2022
	WinCLIP [18]	✗	84.6	85.8	96.8	86.2	94.4	97.0	96.0	88.4	CVPR′2023
	InCTRL [37]	✗	86.3	86.8	89.8	68.1	91.8	95.5	95.2	88.3	CVPR′2024
	AnomalyCLIP [38]	✗	87.8	89.1	84.5	90.8	95.4	97.3	94.9	87.8	ICLR′2024
	PromptAD [11]	✗	88.3	90.0	97.1	85.8	95.7	97.9	96.2	88.5	CVPR′2024
	LogSAD [39]	✗	91.8	92.0	97.8	89.7	96.3	97.6	97.3	93.1	CVPR′2025
	FOUNDAD [19]	✓	93.8	93.3	99.7	98.2	96.9	98.3	97.0	93.2	ICLR′2026
	TCFMAD	✓	94.9	94.5	99.7	98.5	95.7	97.2	97.4	93.6	
4	PatchCore [33]	✓	85.3	87.5	96.8	84.9	88.8	94.5	94.3	84.3	CVPR′2022
	WinCLIP [18]	✗	87.3	88.8	97.2	87.6	95.2	97.3	96.2	89.0	CVPR′2023
	InCTRL [37]	✗	87.8	88.0	90.2	68.8	93.1	96.3	95.8	89.5	CVPR′2024
	AnomalyCLIP [38]	✗	88.8	90.1	85.2	91.4	96.1	97.8	95.5	88.2	ICLR′2024
	PromptAD [11]	✗	89.1	90.8	97.4	86.2	96.6	98.5	96.5	90.5	CVPR′2024
	LogSAD [39]	✗	93.2	93.4	98.1	90.5	96.6	97.7	97.5	93.5	CVPR′2025
	FOUNDAD [19]	✓	94.4	94.0	99.7	98.4	97.1	98.6	97.2	93.6	ICLR′2026
	TCFMAD	✓	95.3	94.7	99.8	98.7	96.1	98.0	96.3	93.0	

view with equal probability. Under each shot setting (1/2/4), the model is trained for 2,000 epochs across all anomaly categories.

For image-level evaluation, top-K average pooling is applied to the patch-level anomaly evidence, with $K = 10$ for MVTec-AD and $K = 6$ for VisA and PKU-Market-PCB. Pixel-level maps are produced by bilinearly upsampling the 32×32 patch-response map to 512×512 with `align_corners=false`; no Gaussian smoothing is used.

For WCB, SAM ViT-H is adapted using 180 densely annotated image–mask pairs, comprising 144 training and 36 validation images, with 30 images available for each of the six PCB defect categories. The image and prompt encoders remain frozen, and only the mask decoder is optimized for 20 epochs.

For PKU-Market-PCB, explicit image-level manifests separate the normal-support, anomalous-patch-support, WCB fine-tuning/validation, and final evaluation partitions.

6) Support-Sample Protocol: For each dataset, support samples are selected once using a fixed support-selection seed of 42. The resulting 1/2/4-shot normal-support manifests are shared by FOUNDAD and TCFMAD, and the class-specific anomalous-patch manifests used by RAS are also kept fixed. Therefore, the reported comparisons control for differences in normal support selection. Unless otherwise stated, each reported value is obtained from one fixed support split rather than averaged over multiple independently sampled splits. A fixed split supports controlled comparison and reproducibility but does not quantify performance variability across alternative support selections.

B. Main Experimental Results

1) Quantitative Comparison With Multi-Class Baselines: Table I reports a quantitative comparison with multi-class-one-model few-shot AD methods on VisA and MVTec-AD under 1/2/4-shot settings. Overall, TCFMAD achieves the best or near-best results across both benchmarks, indicating that enforcing tri-consistency effectively improves both anomaly discrimination and localization under sparse supervision. Compared with classical feature-matching baselines (PatchCore [33] and FastRecon), FM-based methods yield substantially stronger performance, highlighting the benefit of pretrained priors in few-shot manufacturing AD. On VisA, TCFMAD consistently ranks first across all shot settings, improving over FOUNDAD [19] in image-level metrics (e.g., 93.4/94.9/95.3 I-AUROC for 1/2/4-shot) while maintaining very strong localization (P-AUROC around 99.7–99.8 and

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PRO around 98.2–98.7). On MVTec-AD, TCFMAD remains competitive with the best methods and achieves the highest pixel-level AUROC in 1/2-shot (97.3/97.4), with PRO comparable to the top-performing baseline. These results suggest that RAS provides more faithful training signals and CLEA stabilizes anomaly evidence, which is particularly beneficial in low-shot regimes.

2) Quantitative Comparison With One-Class Baseline: Table II compares TCFMAD with representative one-class-one-model few-shot AD baselines on VisA and MVTec-AD under 1/2/4-shot settings. Note that most listed baselines are evaluated in the easier one-class protocol (one model per class), whereas FOUNDAD [19] and TCFMAD follow the more challenging multi-class-one-model generalized setting, where a single model must handle all categories. Despite this stricter protocol, TCFMAD achieves the best performance on VisA across all shots, reaching 93.4/94.9/95.3 I-AUROC for 1/2/4-shot while maintaining near-saturated localization scores (P-AUROC 99.7–99.8 and PRO 98.2–98.7). On MVTec-AD, TCFMAD remains highly competitive, achieving the best pixel-level AUROC in 1/2-shot (97.3/97.4) and PRO comparable to the strongest baselines. Compared with vision-language or one-class prompt-learning approaches (e.g., WinCLIP, PromptAD, LogSAD), TCFMAD and FOUNDAD provide markedly stronger localization, particularly in PRO, suggesting that manifold-projection-style modeling with foundation visual tokens yields more stable pixel-level evidence under few-shot supervision. Importantly, TCFMAD consistently improves upon FOUNDAD on VisA under all shot settings, indicating that enhancing synthetic-to-real anomaly consistency and cross-layer evidence stability benefits generalization on the more diverse VisA benchmark. Overall, these results demonstrate that TCFMAD can outperform strong one-class baselines even under a harder multi-class setting, supporting its practicality for real manufacturing scenarios where deploying one model per category is often infeasible.

3) Quantitative Comparison in PKU-Market-PCB: Table III reports per-class results on PKU-Market-PCB under 1/2/4-shot settings, comparing FOUNDAD (BS) with TCFMAD under the same fixed WCB-refined pseudo-segmentation masks. TCFMAD achieves higher image-level discrimination across all shots (mean I-AUROC/AUPR: 71.3/72.6, 69.5/72.5, and 71.0/74.5 for 1/2/4-shot) than FOUNDAD (61.7/60.3, 59.7/61.0, and 60.3/61.3). Under the shared pseudo-mask protocol, TCFMAD also obtains higher mean P-AUROC (89.7/89.5/89.9 vs. 87.6/88.1/89.8) and PRO (72.0/73.8/74.0 vs. 63.5/67.8/72.0). Because both methods use identical fixed reference masks, these differences reflect the relative spatial alignment of their anomaly maps under the common pseudo-mask assessment protocol rather than an improvement produced by WCB. The absolute PKU-Market-PCB localization values should nevertheless be interpreted separately from the human-ground-truth results on MVTec-AD and VisA.

TABLE III

PER-CLASS RESULTS (%) ON PKU-MARKET-PCB UNDER DIFFERENT 1/2/4-SHOT SETTINGS. FOUNDAD (BS) AND TCFMAD (OUR) ARE EVALUATED USING THE SAME FIXED SET OF WCB-REFINED PSEUDO-SEGMENTATION MASKS. THESE MASKS ARE GENERATED ONCE BEFORE DETECTOR EVALUATION AND REMAIN INDEPENDENT OF THE PREDICTIONS OF EITHER METHOD. THE REPORTED P-AUROC AND PRO VALUES CONSTITUTE PSEUDO-MASK-BASED LOCALIZATION ASSESSMENT AND ARE NOT DIRECTLY EQUIVALENT TO THE HUMAN-GROUND-TRUTH LOCALIZATION RESULTS ON MVTEC-AD AND VISA. THE BETTER RESULT IN EACH COMPARISON PAIR IS HIGHLIGHTED IN BLUE

Class	Metric	1-shot		2-shot		4-shot	
		BS	Our	BS	Our	BS	Our
miss. hole	I-AUROC	51.5	62.9	58.4	66.1	55.8	69.1
	I-AUPR	53.2	61.5	55.8	59.9	54.8	65.4
	P-AUROC	91.5	95.3	94.9	95.8	95.1	95.8
	PRO	74.5	86.4	86.4	89.3	87.1	90.0
mouse_bite	I-AUROC	65.1	81.3	60.1	75.7	58.9	77.7
	I-AUPR	63.5	82.8	61.0	80.1	61.4	83.6
	P-AUROC	89.6	92.1	89.7	90.5	91.2	90.8
	PRO	65.9	76.0	67.0	72.1	70.8	70.7
open_circ.	I-AUROC	67.1	68.3	57.7	66.7	62.3	69.2
	I-AUPR	70.2	68.3	63.4	72.3	67.2	75.8
	P-AUROC	89.3	89.6	90.3	92.2	93.0	92.9
	PRO	66.8	70.4	72.0	76.8	78.7	78.3
short	I-AUROC	63.1	67.5	61.6	66.5	62.4	64.1
	I-AUPR	60.1	69.4	64.5	70.6	59.9	66.4
	P-AUROC	88.3	89.5	87.9	89.1	89.6	89.0
	PRO	68.3	75.4	74.0	81.1	79.0	77.9
spur	I-AUROC	64.5	75.7	60.1	68.6	60.6	69.8
	I-AUPR	60.0	78.4	61.4	75.4	61.3	75.2
	P-AUROC	79.1	81.7	78.3	80.1	80.2	80.0
	PRO	43.1	54.7	44.2	55.3	49.1	53.8
spur. cop.	I-AUROC	58.7	71.9	60.6	73.1	61.7	76.3
	I-AUPR	54.9	75.3	59.9	77.0	63.3	80.5
	P-AUROC	87.7	89.8	87.7	89.6	89.5	90.8
	PRO	62.0	69.0	63.2	68.1	67.4	73.0
Mean	I-AUROC	61.7	71.3	59.7	69.5	60.3	71.0
	I-AUPR	60.3	72.6	61.0	72.5	61.3	74.5
	P-AUROC	87.6	89.7	88.1	89.5	89.8	89.9
	PRO	63.5	72.0	67.8	73.8	72.0	74.0

C. Ablation Study

1) Effectiveness of Detector-Side Components: Table IV presents an ablation study of the detector-side components RAS and CLEA on VisA, MVTec-AD, and PKU-Market-PCB. WCB is excluded from this component ablation because it changes only the evaluation labels and does not modify the detector. Removing either RAS or CLEA leads to performance degradation, confirming that realistic anomaly synthesis and

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TABLE IV

ABLATION STUDY OF THE DETECTOR-SIDE COMPONENTS RAS AND CLEA ON VISA, MVTEC-AD, AND PKU-MARKET-PCB. MVTEC-AD AND VISA LOCALIZATION METRICS USE HUMAN-ANNOTATED MASKS, WHEREAS PKU-MARKET-PCB USES FIXED WCB-REFINED PSEUDO MASKS; THE CORRESPONDING ABSOLUTE LOCALIZATION VALUES ARE NOT DIRECTLY EQUIVALENT ACROSS THESE ANNOTATION PROTOCOLS. WCB IS EXCLUDED BECAUSE IT REFINES THE EVALUATION LABELS WITHOUT MODIFYING THE ANOMALY DETECTOR

Method	VisA				MVTec-AD				PKU-Market-PCB			
	I-AUROC	AUPR	P-AUROC	PRO	I-AUROC	AUPR	P-AUROC	PRO	I-AUROC	AUPR	P-AUROC	PRO
FOUNDAD	92.6	92.0	99.7	98.0	96.1	97.9	96.8	92.8	61.7	60.3	87.6	63.5
TCFMAD	93.4	93.3	99.7	98.2	95.8	97.0	97.3	93.6	71.3	72.6	89.7	72.0
w/o RAS	93.0	92.8	99.7	98.1	95.5	96.1	96.6	93.3	70.9	70.3	89.2	71.7
w/o CLEA	92.7	92.5	99.7	98.0	95.3	96.4	96.5	93.0	65.5	66.3	88.6	71.6

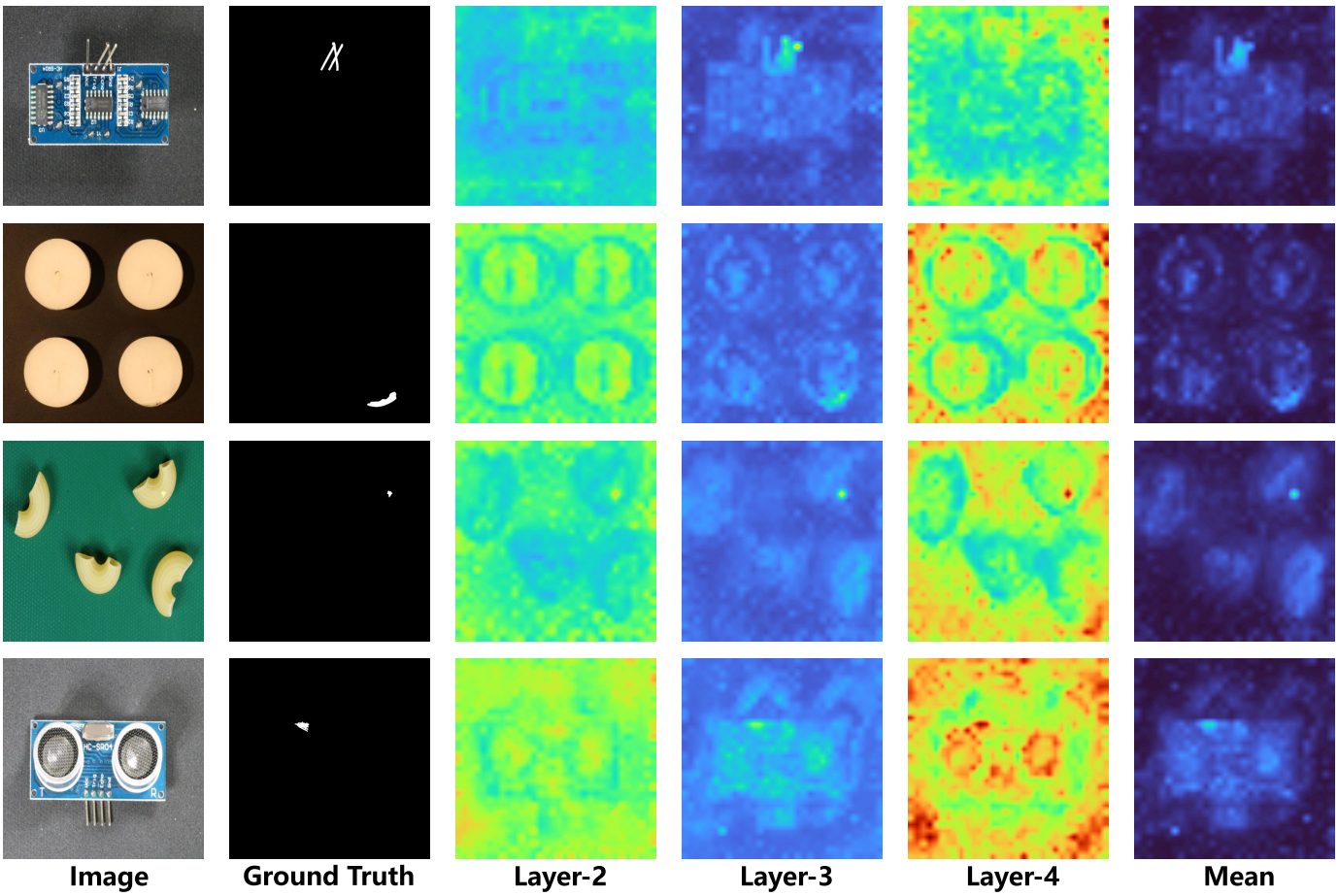


Fig. 4. Layer-wise analysis of anomaly evidence from different DINOv3 representation depths. From left to right: input image, ground-truth mask, evidence maps from the second-, third-, and fourth-last Transformer layers, and their arithmetic-mean fusion. Individual layers provide complementary responses but may produce isolated high-response regions outside the annotated defect. Mean fusion attenuates these unsupported hotspots while preserving strong evidence within the anomalous regions. All evidence maps within each row share the same color scale. The final column is a simple mean-fusion reference rather than the output of the complete learned-fusion CLEA model.

cross-layer evidence stabilization both contribute to few-shot anomaly detection and localization. On MVTec-AD, removing RAS decreases AUPR from 97.0 to 96.1 and P-AUROC from 97.3 to 96.6. Removing CLEA also degrades performance on both public benchmarks. The detector-side effects are more evident on PKU-Market-PCB: removing CLEA decreases I-AUROC from 71.3 to 65.5 and AUPR from 72.6 to 66.3, while removing RAS decreases P-AUROC from 89.7 to 89.2 and PRO from 72.0 to 71.7. These results support the respective roles of RAS and CLEA under a common evaluation protocol.

2) Layer-Wise Anomaly-Evidence Analysis: Figure 4 visualizes anomaly evidence from the second-, third-, and fourth-last DINOv3 layers and their arithmetic-mean fusion. The individual layers exhibit clearly different response patterns, confirming that different representation depths capture complementary but potentially unstable anomaly cues. Across the four examples, an individual layer exhibits a pronounced

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TABLE V

EFFECT OF REFERENCE-MASK GRANULARITY ON THE LOCALIZATION ASSESSMENT OF FIXED DETECTOR PREDICTIONS. FOR EACH METHOD, THE ANOMALY MAPS AND IMAGE-LEVEL SCORES REMAIN UNCHANGED; ONLY THE REFERENCE ANNOTATIONS USED TO COMPUTE P-AUROC AND PRO DIFFER. BOX-MASK EVALUATION USES THE ORIGINAL COARSE BOUNDING BOXES, WHEREAS REFINED-MASK EVALUATION USES THE SAME FIXED WCB-GENERATED PSEUDO MASKS FOR ALL DETECTORS

Method	Shot	Evaluation Mask	Metrics			
			I-AUROC	AUPR	P-AUROC	PRO
FOUNDAD	1-shot	Box mask	61.7	60.3	78.1	42.0
		WCB mask	61.7	60.3	87.6	63.5
	2-shot	Box mask	59.7	61.0	77.9	46.3
		WCB mask	59.7	61.0	88.1	67.8
	4-shot	Box mask	60.3	61.3	78.6	46.8
		WCB mask	60.3	61.3	89.8	72.0
TCFMAD	1-shot	Box mask	71.3	72.6	79.7	49.4
		WCB mask	71.3	72.6	89.7	72.0
	2-shot	Box mask	69.5	72.5	78.3	49.6
		WCB mask	69.5	72.5	89.5	73.8
	4-shot	Box mask	71.0	74.5	79.1	49.8
		WCB mask	71.0	74.5	89.9	74.0

response outside the annotated defect that is not supported by the other two layers. These cases include responses associated with background textures, object boundaries, printed patterns, and normal structural regions. Mean fusion attenuates these isolated activations while retaining a strong response over the true anomalous region, including small and localized defects. This observation indicates that cross-layer aggregation can improve evidence reliability by emphasizing responses supported across representation depths and reducing layer-specific false positives. Together with the quantitative results obtained by removing CLEA in Table IV, the visualization provides complementary qualitative evidence for the proposed cross-layer mechanism.

3) Effect of Evaluation-Mask Granularity: Table V compares two annotation protocols for evaluating the same fixed anomaly predictions on PKU-Market-PCB. Since the image-level labels and detector outputs remain unchanged, I-AUROC and AUPR are identical under both protocols. The differences in P-AUROC and PRO arise entirely from the spatial granularity of the reference annotations.

The coarse box masks label every pixel within each bounding box as anomalous and therefore include substantial normal regions. This makes the resulting localization metrics sensitive to box extent and annotation tightness. By contrast, WCB-refined pseudo masks provide more compact reference regions and enable a more fine-grained assessment of whether the predicted anomaly evidence is spatially aligned with the defect. Accordingly, the metric differences in Table V should not be interpreted as detector improvements caused by WCB. Rather, they reveal the assessment bias introduced by coarse bounding boxes and motivate refined reference annotations when human pixel-level masks are unavailable.

For fair comparison, a single fixed set of WCB-refined pseudo masks is shared by FOUNDAD and TCFMAD. Under this identical protocol, TCFMAD obtains stronger localization results than FOUNDAD. Since the reference masks are the same for both methods, the difference reflects their relative spatial alignment under the common pseudo-mask assessment protocol. These PKU-Market-PCB values remain separate from the human-ground-truth localization results on MVTec-AD and VisA.

TABLE VI

SENSITIVITY ANALYSIS OF THE HYPERPARAMETER λ. METRICS INCLUDE IMAGE-LEVEL AUROC (%), IMAGE-LEVEL AUPR (%), PIXEL-LEVEL AUROC (P-AUROC, %), AND PRO (%). THE BEST RESULT IN EACH METRIC IS HIGHLIGHTED IN BLUE

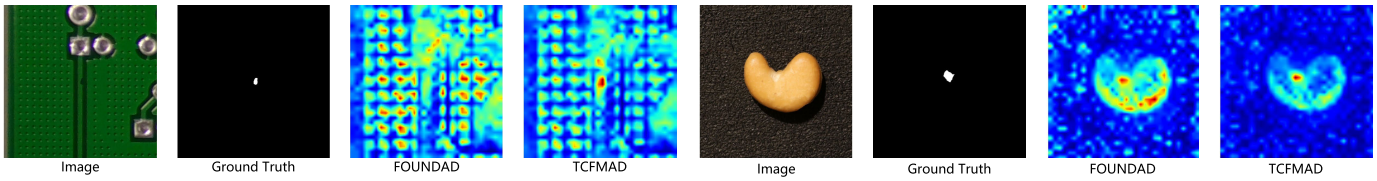
λ	I-AUROC	I-AUPR	P-AUROC	PRO
0.1	94.91	94.53	99.73	98.47
0.2	94.04	94.17	99.73	98.30
0.3	94.08	94.11	99.73	98.36
0.4	94.08	94.29	99.73	98.39
0.5	94.10	94.29	99.74	98.38
1.0	94.09	94.13	99.73	98.35

4) Sensitivity to the Consistency Weight in CLEA: Table VI reports a sensitivity study on the consistency loss coefficient λ in CLEA. Overall, performance is stable across a wide range of λ values, indicating that the proposed cross-layer agreement is not overly sensitive to this hyperparameter. The best image-level discrimination is achieved at a small coefficient, with $\lambda = 0.1$ yielding the highest I-AUROC (94.91) and I-AUPR (94.53) as well as the best PRO (98.47). Pixel-level AUROC remains nearly saturated and varies marginally across settings (99.73–99.74), suggesting that localization performance is largely preserved once cross-layer fusion is enabled, while λ mainly controls the strength of agreement regularization. These results support the robustness of CLEA and validate that a modest consistency weight (e.g., $\lambda = 0.1$) provides a good trade-off between stable anomaly evidence agreement and overall detection performance.

5) Runtime and Resource Analysis: Table VII reports online inference, projected training cost, peak allocated GPU memory, and offline WCB preprocessing. FOUNDAD and TCFMAD are profiled under the same online protocol on one NVIDIA GeForce RTX 3090 using 512×512 inputs, batch size 1, FP32 precision, 50 warm-up iterations, and 400 synchronized timed images. FOUNDAD requires 24.770 ms/image (40.371 FPS) and 500.047 MiB peak allocated inference memory, whereas TCFMAD requires 25.161 ms/image (39.744 FPS) and 509.118 MiB. Thus, TCFMAD introduces only 1.58% latency overhead and approximately 9.071 MiB additional inference memory. The exact parameter counts are 96.908544M for FOUNDAD and 96.913152M for TCFMAD, so the learnable cross-layer fusion adds only 4,608 parameters.

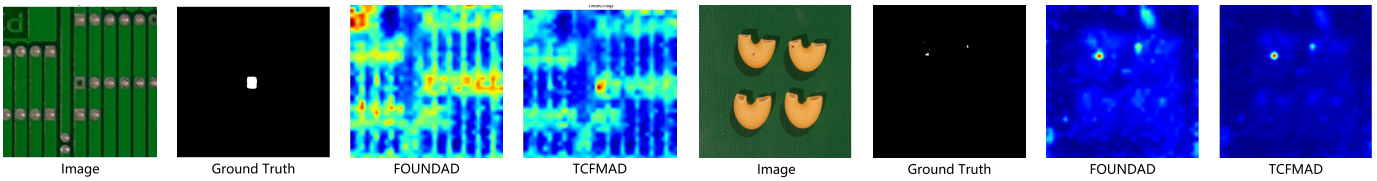
For training, we use the same normal-support manifest, batch size, and optimization protocol for the two methods. After 10 warm-up iterations, 100 steady-state updates are measured and projected to the common 2,000-epoch schedule described in Section IV-A. The resulting estimated training times are 5.667 h for FOUNDAD and 2.091 h for TCFMAD, with peak allocated training memory of 5,158.8 and 16,087.3 MiB, respectively. These times are estimates from

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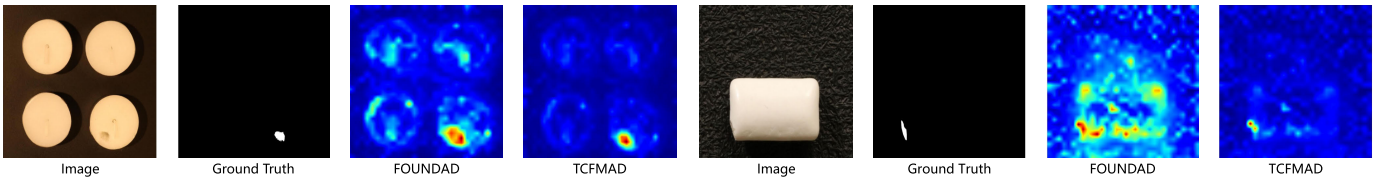
The baseline highlights the background and misses the true cut anomaly on the PCB.

The baseline misses the crack in the cashew and falsely highlights its edges.



The baseline misses the missing-hole anomaly and highlights the letter d instead.

The overlapping region is incorrectly highlighted as anomalous.



The baseline highlights the defective candle but also falsely activates normal ones.

The baseline highlights all corners of the chewing-gum samples as anomalous.

Fig. 5. Qualitative comparison of anomaly evidence across various manufacturing scenarios. The baseline FOUNDAD [19] often exhibits spurious activations and localization drift, such as highlighting background regions or text in PCB images and falsely activating corners of food products. In contrast, TCFMAD produces more compact and spatially aligned evidence that better focuses on true defects.

TABLE VII

RUNTIME AND RESOURCE ANALYSIS OF FOUNDAD, TCFMAD, AND OFFLINE WCB PREPROCESSING. ONLINE INFERENCE IS MEASURED ON ONE NVIDIA GEFORCE RTX 3090 WITH 512 × 512 INPUTS, BATCH SIZE 1, FP32 PRECISION, 50 WARM-UP ITERATIONS, AND 400 SYNCHRONIZED TIMED IMAGES. TRAINING TIME IS PROJECTED FROM MEASURED STEADY-STATE UPDATES RATHER THAN A COMPLETE WALL-CLOCK RUN

Method	Params (M)	Latency (ms)	FPS	Memory (MiB)
FOUNDAD	96.908544	24.770	40.371	500.047
TCFMAD	96.913152	25.161	39.744	509.118

Method	Est. 2,000-epoch time (h)	Training memory (MiB)
FOUNDAD	5.667	5,158.8
TCFMAD	2.091	16,087.3

Offline WCB stage	Workload	Total (s)	Latency (ms/image)	Throughput (images/s)
SAM mask-decoder fine-tuning	20 epochs	533.054	–	–
Mask generation (SAM ViT-H)	590 images	278.411	471.884	2.119
Complete WCB pipeline	–	811.465	–	–

Training times are estimated from 100 steady-state updates after 10 warm-up iterations. All 590 WCB-generated masks were nonempty. WCB is offline and is excluded from online TCFMAD latency and FPS.

measured iteration durations rather than complete wall-clock training runs. TCFMAD requires more model-side memory because CLEA performs multi-layer reconstruction and consistency optimization. Its lower measured total iteration time is attributable to RAS being substantially faster than the original CutPaste synthesis used by FOUNDAD, rather than to lower CLEA computation.

WCB is executed offline and is excluded from online latency and FPS. The archived 20-epoch SAM mask-decoder fine-tuning requires 533.054 s. Using the fine-tuned SAM ViT-H checkpoint, mask generation for all 590 PKU-Market-PCB images requires 278.411 s, or 471.884 ms/image at 2.119 images/s. The complete offline WCB pipeline therefore requires 811.465 s, and all generated masks are nonempty.

D. Anomaly Detection Visualization

In Fig. 5, qualitative results further demonstrate the superiority of TCFMAD in anomaly evidence localization. Compared with FOUNDAD, which often exhibits spurious activations on background, text, edges, or normal instances, TCFMAD produces more compact, spatially aligned, and semantically accurate anomaly evidence. Specifically, TCFMAD is able to focus on the true defective regions such as PCB cuts, missing holes, and cashew cracks, while suppressing false responses on irrelevant structures like board text, object boundaries, and overlapping regions. It also shows stronger robustness in cluttered or multi-instance scenes, where the baseline tends to activate normal candles or the corners of chewing-gum samples as anomalies. These visual comparisons indicate that TCFMAD not only improves anomaly discrimination but also significantly enhances localization reliability by reducing localization drift and false-positive hotspots.

V. CONCLUSION

In this paper, we tackle foundation-model-based few-shot anomaly detection for manufacturing by addressing triple inconsistencies before training, during training, and during evaluation. On the detector side, RAS reduces the synthetic-to-real gap by generating realistic anomalies from class-consistent anomalous patches, while CLEA stabilizes patch-token anomaly evidence across multiple DINOv3 layers

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through consistency regularization. WCB is distinct from these detector-side contributions: it does not modify anomaly detection or localization predictions, but instead reduces box-induced localization-assessment bias by refining weak bounding boxes into one fixed set of fine-grained pseudo masks shared by all compared methods. The PKU-Market-PCB localization results are therefore interpreted under this pseudo-mask protocol and separately from the human-ground-truth results on MVTec-AD and VisA. Under these clearly separated roles and evaluation protocols, TCFMAD achieves strong anomaly discrimination and localization performance across the evaluated manufacturing datasets.

The current evaluation follows one fixed support split for each few-shot setting. Although this protocol enables controlled and reproducible comparisons, it does not quantify performance variability under different support selections. Evaluating support-sample sensitivity over multiple independent splits remains an important direction for future work.

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